

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	B . Nandini	Reg. No.:	192525137
Date:	04-08-2026	Duration:	60 minutes

Code of Conduct

I B . Nandini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B . Nandini

Annexure A – Scenario-Based Requirement Analysis

Agriculture Equipment Rental and Farm Resource Management Platform

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.

Problem Description

Many farmers cannot afford expensive agricultural machinery such as tractors, harvesters, seeders, and rotavators. Most equipment remains unused after seasonal farming. Farmers also struggle to maintain records of farm resources like seeds, fertilizers, pesticides, labor, and expenses.

The proposed platform provides an online marketplace where equipment owners can rent machinery to farmers while also helping farmers manage their farm resources efficiently.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain

Agriculture / AgriTech / Equipment Rental Management System

Existing Challenges

- High cost of agricultural equipment
- Lack of equipment availability information
- Manual booking process
- Poor farm resource management
- No online payment facility
- Lack of booking transparency
- Equipment scheduling conflicts

System Objectives

- Digitize equipment rental
- Improve equipment utilization
- Reduce farming costs
- Manage farm resources digitally
- Enable online booking and payment
- Generate reports for administrators

Stakeholders

Stakeholder	Responsibility
Farmers	Rent equipment and manage resources
Equipment Owners	List and manage equipment
Administrator	Manage platform
Delivery Staff	Deliver equipment
Payment Gateway	Process payments
Agricultural Suppliers	Provide farming products

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

FR1

User Registration

FR2

User Login

FR3

Farmer Dashboard

FR4

Equipment Owner Dashboard

FR5

Equipment Listing

FR6

Equipment Search

FR7

Equipment Booking

FR8

Booking Cancellation

FR9

Online Payment

FR10

Payment History

FR11

Farm Resource Management

FR12

Expense Management

FR13

Equipment Availability Tracking

FR14

Notifications

FR15

Reports Generation

FR16

User Profile Management

FR17

Feedback and Ratings

FR18

Admin Management

- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Non-Functional Requirements

Performance

- System should load within 3 seconds.
- Support 1000+ users simultaneously.

Security

- Secure login authentication
- Encrypted passwords
- Secure online payment
- Role-based access control

Reliability

- 99% uptime
- Automatic backup

Availability

- Available 24×7

Usability

- Simple user interface
- Mobile-friendly

Scalability

- Easy to add new equipment categories
- Support increasing users

Maintainability

- Modular architecture
- Easy software updates

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

- Farmer
- Equipment Owner
- Administrator

Secondary Actors

- Delivery Staff
 - Payment Gateway
 - SMS/Email Notification Service
- List the major use cases associated with each actor.

Use Cases

Farmer

- Register
- Login
- Search Equipment
- View Equipment
- Book Equipment
- Cancel Booking
- Pay Online
- Manage Farm Resources
- View Booking History

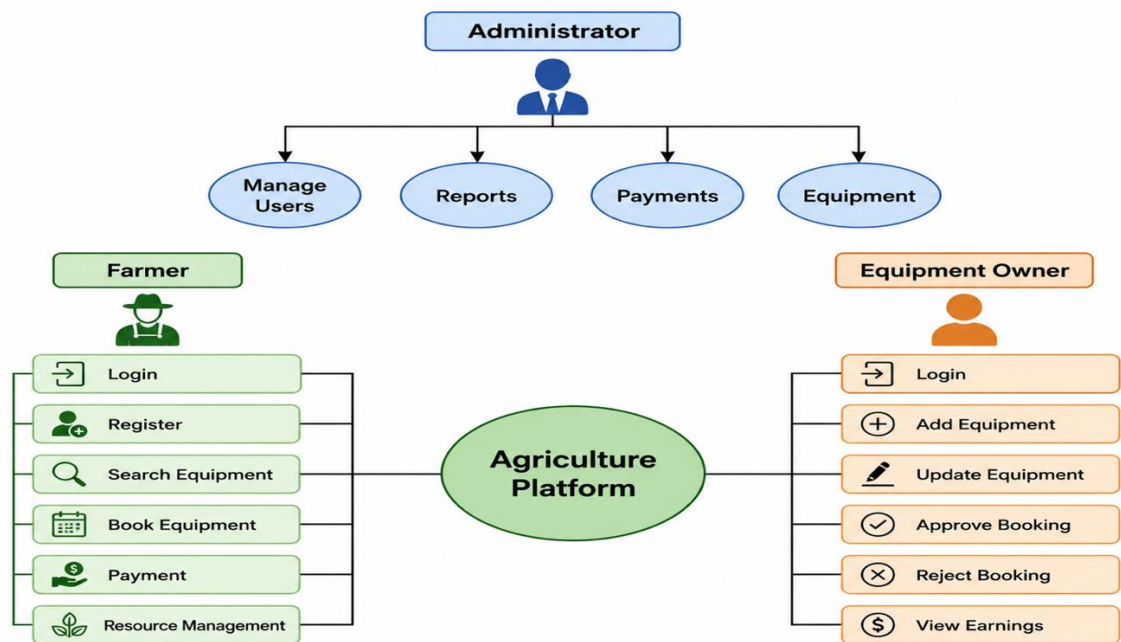
Equipment Owner

- Register
- Login
- Add Equipment
- Update Equipment
- Delete Equipment
- Approve Booking
- Reject Booking
- View Earnings

Administrator

- Manage Users
- Verify Equipment
- View Reports
- Manage Payments
- Generate Reports

- Draw a Use Case Diagram representing the interactions between actors and the system.



4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

- Introduction

The Agriculture Equipment Rental and Farm Resource Management Platform enables farmers to rent agricultural machinery online while efficiently managing farm resources such as seeds, fertilizers, labor, and expenses.

- Problem Description

Small-scale farmers often face financial difficulties purchasing costly machinery. Manual rental systems lack transparency and efficiency. Farm resource records are typically maintained manually, leading to poor planning and increased operational costs.

- Scope of the System

The system allows:

- Equipment rental
- Farm resource management
- Online payment
- Booking management
- Report generation
- User management

- Functional Requirements

- User Registration
- Login
- Equipment Listing
- Equipment Search
- Booking
- Payment
- Resource Management
- Reports

- Non-Functional Requirements

- Security
- Performance
- Reliability
- Availability
- Scalability
- Maintainability

- Use Case Descriptions

Use Case: Book Equipment

Actor: Farmer

Precondition

- User logged in

Steps

1. Search equipment
2. Select equipment
3. Choose rental date
4. Confirm booking
5. Pay online

Postcondition

Booking confirmed.

- Data Requirements

User

- User ID
- Name
- Phone
- Email
- Address
- Password

Equipment

- Equipment ID
- Equipment Name
- Owner Name
- Rental Price
- Availability

Booking

- Booking ID
- Farmer ID
- Equipment ID
- Booking Date
- Rental Period
- Payment Status

Farm Resource

- Seed Name
- Fertilizer
- Labor
- Expenses
- Quantity

- External Interface Requirements

User Interface

- Web Application
- Mobile Responsive

Hardware

- Computer
- Smartphone

Software

- Web Browser
- Database Server

Communication

- Internet Connection
 - System Features
- Secure Login
- Equipment Rental
- Resource Tracking
- Online Payment
- Notifications
- Reports
- Booking History

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

Assumptions

- Internet connection is available.
- Users possess smartphones or computers.
- Equipment owners provide accurate information.
- Payment gateway functions properly.
- GPS/location services are available if required.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Technical

- Requires internet connectivity.

- Payment API integration.

Operational

- Users need basic digital literacy.

Economic

- Development budget is limited.

Legal

- User data must comply with privacy regulations.
- Secure payment compliance.

Time

- Project completion within one semester.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

Completeness

- ✓ All stakeholders identified.
- ✓ Functional requirements covered.
- ✓ Non-functional requirements covered.
- ✓ Actors identified.
- ✓ Use cases included.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

Consistency

- No conflicting requirements.
- Booking depends on equipment availability.
- Payments occur after booking confirmation.

Ambiguity Check

All requirements are written clearly.

Example:

✓ "User shall book available equipment."

instead of

✗ "User can book equipment."

Missing Requirements Added

- Notification System
 - Equipment Ratings
 - Booking History
 - Payment History
 - Admin Dashboard
 - Farm Expense Tracking
-
- Suggest improvements to enhance the quality and completeness of the requirement specification.
-
- Add AI-based equipment recommendations.
 - Integrate weather forecasting.
 - Enable GPS tracking for equipment delivery.
 - Support multiple regional languages.
 - Develop Android and iOS mobile applications.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25